

Neural-Symbolic Verification of LLM-Generated Scientific Claims: When Does a Calculation Apply?

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Program Officer: Dr. Richard D. (Doug) Riecken
Proposer: Xinya Du
Institution: The University of Texas at Dallas
Email: Xinya.Du@UTDallas.edu
Phone: +1 (972) 883-2634
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Proposed Research and Motivation

LLM-generated scientific claims can be incorrect even when their supporting calculations are mathematically valid. Suppose a language model reads that gas in a sealed, rigid vessel is heated from 300 K to 600 K and concludes that its volume doubles. A calculator confirms the temperature ratio of two. Yet the volume cannot change in a rigid vessel: under the ideal-gas model with fixed gas amount, pressure doubles instead. The volume–temperature relation would require constant pressure [11]. The arithmetic is correct, but the model has applied the wrong scientific relation. In harder cases, the deciding condition appears in another passage or depends on an earlier check. We will study these cases in Task 2.

This project asks: **How can a system verify a scientific claim by establishing that the conditions required by its supporting calculations hold, and preserving those conditions across different reasoning methods?** We will develop a verifier that connects evidence in scientific text to quantitative calculations and logical checks. Each component will return both its result and the conditions under which that result is justified. The system will identify unsupported conditions, seek relevant evidence, and revise a claim only when the revised reasoning is supported.

Our hypothesis is that making these conditions explicit and checking them as results pass between tools will reduce false acceptance of scientific claims compared with running tools and using a learned score to judge correctness. The benefit may disappear if extracting conditions introduces more errors than it detects, or if conservative checks reject too many valid claims. We will test both possibilities at the same computational cost and acceptance rate. The initial scope is quantitative claims about elementary thermodynamic and mechanical systems, expressed in short scientific passages and tables. SciBench identifies difficulty recognizing assumptions as a source of scientific reasoning errors, and MathTrap finds failures to combine known facts with conditions that invalidate standard solutions [15, 18]. We will test whether current models make similar errors.

Relation to current work. Classical compositional modeling used hand-written domain theories to represent operating assumptions and check model behavior [5]. FacTool and AutoVerifier organize tool-based checking and technical evidence [1, 4]; ProgramFC decomposes claims into programs of specialized checks [13]. Logic-LM and SatLM connect language-generated specifications to symbolic solvers [12, 17]. Neuro-symbolic PRM already distinguishes symbolic validity from semantic grounding and scores verifier-passing steps [19]. We will study how to extract these conditions from scientific text, detect missing or invented restrictions, and keep the conditions attached to results as tools exchange them. A companion CCMI white paper studies uncertain document evidence and argument revision; this project studies scientific principles, their applicability conditions, units, and variable bindings across calculation and constraint solving.

Research Hypothesis and Technical Approach

The three tasks use a common record linking each scientific claim to its evidence, principles, and required conditions. Task 1 builds the record, Task 2 checks it, and Task 3 uses the errors found to guide correction.

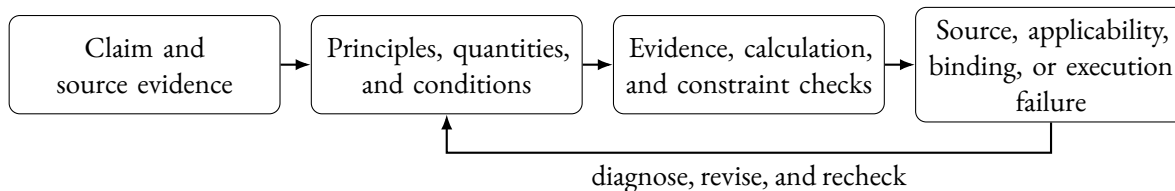


Figure 1: The system records the conditions needed for each check and uses failed checks to guide further retrieval and correction.

Task 1: Recover Scientific Principles and Their Conditions

A useful representation must preserve more than an equation. We will construct a compact record containing the principle and source passage; variables, units, entities, and measurement conditions; the executable relation; and its applicability conditions. Observed facts, modeling assumptions, and derived results will be distinguished. In the vessel example, “sealed” supports fixed gas amount and “rigid” supports fixed volume; neither supports constant pressure. The ideal-gas approximation must also be stated or justified by the supplied context.

Starting from a claim, the system will retrieve a small set of relevant scientific sources and extract only the principles needed for verification. A language model will propose the records. Type and unit checks will detect structural errors, while evidence checks will assess whether the conditions and assignments of quantities to variables match the source. A missing condition will remain unknown rather than be silently assumed true. Conflicting sources will retain their separate conditions instead of being merged into one apparently supported rule.

This builds on our work on document-level template filling [3] and the SciEvent benchmark for scientific event extraction [2]. The new challenge is recovering the restrictions under which extracted knowledge can be used. We will compare extracting equations alone, extracting equations with nearby qualifiers, and retrieving additional context to resolve missing conditions. Independent annotations will measure both omitted conditions and invented restrictions. Using complete records supplied by annotators will let us separate extraction errors from later reasoning errors.

Research question. What information must a principle record contain, and when does retrieving more context improve verification enough to justify the cost?

Task 2: Coordinate Checks Without Losing Their Assumptions

Minsky and colleagues argued that combining specialized reasoning methods requires knowing when each method is suitable and enabling the exchange of partial results [10]. We will study how to keep track of conditions extracted from scientific text as evidence checkers, calculators, and constraint solvers exchange results.

We will develop a planner that turns the claim into a finite set of required checks. Three components will share the principle records: an evidence checker links facts and conditions to passages; a calculation engine evaluates equations and units; and a constraint solver checks compatibility among conditions. We will use a satisfiability modulo theories (SMT) solver to check numerical and logical constraints. This builds on our work on interpreting the logic of natural-language explanations [6].

The planner will work backward from the target quantity or relation. It will retrieve candidate principles, identify their required inputs and conditions, and assign each requirement to a component. Checks that use another component’s output will also inherit the conditions attached to that output. For example, a calculation of doubled volume cannot be accepted merely because its arithmetic passes: the evidence checker must establish constant pressure, and the constraint checker must reconcile that requirement with the rigid-vessel description. Unresolved requirements trigger targeted retrieval or an unresolved verdict; they cannot be satisfied by the claim under examination itself.

Each component will return a *verification record*: its principle, source, assignments to variables, operation, result, and conditions marked as supported, contradicted, or unresolved. We will check both **whether the calculation ran correctly** and **whether it applies to the evidence**. Running successfully is not enough to establish scientific correctness. Evidence judgments remain fallible even when calculations and encoded constraints are checked exactly.

The main experiment will test whether exchanging conditions improves decisions compared with giving the same evidence and tools to a neural planner or a verifier-first process reward model. We will vary whether a relevant condition appears beside its equation, in another passage, or only as the conclusion of an earlier check. Even when intermediate results pass their individual checks, they may depend on incompatible conditions or refer to different objects or times.

For finite plans, we will define acceptance rules that keep all required conditions attached to results. Combining results must not turn an unresolved condition into a supported one. Correctness under stated conditions is a design requirement. We will study what source and context information must accompany intermediate results, and how errors in exchanging that information lead the system to accept invalid claims. We will analyze restricted compositions and test the limits experimentally, without promising formal guarantees for unrestricted scientific text.

Research question. What must checkers tell one another before their results can support a scientific conclusion?

Task 3: Diagnose and Correct the Failed Scientific Inference

Verification records will identify whether an error comes from unsupported evidence, a principle that does not apply, a quantity assigned to the wrong variable, or a failed calculation. A model will propose a correction using the failed requirement and its evidence. In the vessel example, the corrected statement says that volume remains fixed; pressure doubles under the stated ideal-gas and fixed-amount assumptions. If those assumptions cannot be established, the output will explain what remains unresolved rather than invent evidence.

A revised claim will be accepted only if the failed checks are resolved without invalidating components that previously passed. The system will recheck the revised claim and any dependent results. It must preserve the original question, objects, and operating setting; deleting a difficult condition or replacing a specific assertion with a vague one will not count as repair. Corrections that legitimately narrow a claim’s scope will be labeled separately from corrections within its original scope.

Our FaithScore method breaks outputs into claims for factual assessment, and FG-PRM detects errors within mathematical reasoning steps [7, 8]. We will extend this approach to errors in applying principles and assigning quantities to variables. We will first use prompted correction with external feedback, comparing feedback that names the error type with solver-error feedback and a complete rerun of verification at the same cost. Training on verified counterfactual examples will be optional, pursued only if the initial system shows a clear need. Such examples will not supply the final evaluation labels.

Research question. Does identifying the failed requirement help correct a claim without changing its intended scope or introducing new errors?

Evaluation and Measures of Success

The first comparison uses paired cases that change one applicability condition while preserving the numerical calculation. Evaluation will combine controlled cases with document-based claims. Annotators with relevant scientific training will independently construct cases from public scientific sources, including held-out principles and compositions, with valid claims and matched applicability or binding violations. The source documents will be available to every system; reference encodings and labels will remain hidden. Splits will separate principles and source documents, not merely paraphrases. Human-authored and automatically perturbed results will be reported separately. SciFact and SCITAB provide complementary tests of textual evidence and numerical table reasoning [9, 14]; neither alone measures the proposed applicability problem.

The primary outcome is the fraction of accepted claims that are invalid, at matched coverage (the fraction accepted) and total cost, including retrieval, model calls, execution, and retries. We will also report rejection of valid claims, unresolved cases, condition-extraction accuracy, error localization, and corrections that preserve the claim’s scope without introducing new errors. Labels will come from independent evidence and step annotations, not the system’s own judgments. Cases without sufficient

information will have an unresolved reference outcome. Reporting acceptance rates will show whether a method appears more reliable simply because it accepts fewer claims.

Baselines will include retrieval plus an LLM judge, a ProgramFC-style planner adapted to the same scientific tools without explicit condition exchange [13], verifier-first process scoring, and symbolic checking supplied with human-validated records. Ablations will remove condition extraction, condition exchange between components, or feedback that identifies the type of error. We will test whether the method accepts fewer invalid claims than the strongest tool-augmented baseline at the same coverage and cost on held-out principles. The target margin and sample size will be fixed before testing. We will report uncertainty intervals and failures from extraction or overly restrictive checking.

Potential Impact on DAF and DoW Capabilities

The Science of Information, Computation, Learning, and Fusion portfolio calls for connecting data-driven extraction with knowledge-driven reasoning and integrating diverse reasoning components. We address these objectives by studying what the components must exchange to support a scientific conclusion.

Motivating settings include reviewing AI-generated engineering analyses and checking whether measurements support claims under documented operating conditions. An analyst would receive the relevant source, calculation, conditions, and a specific explanation of any unsupported step. This could reduce repeated manual checking and make incorrect uses of otherwise valid principles easier to detect. The project will use public sources and unclassified document tasks; it requires neither an operational deployment nor access to restricted mission data.

Team and Development Plan

PI Xinya Du will lead the work at UT Dallas with graduate and undergraduate researchers. Our prior research provides a foundation in scientific information extraction, logical reasoning, detailed evaluation, and scientific hypothesis discovery [2, 6, 8, 16]. No external collaborators or subawards are planned. The estimated three-year budget is \$450,000 including indirect costs, supporting personnel, computation, independent annotation, and dissemination.

Year 1 will deliver an end-to-end prototype evaluated on independently annotated thermodynamic cases. **Year 2** will study condition extraction and composition, using controlled failures to refine the interfaces and acceptance rules. **Year 3** will evaluate correction and transfer to held-out mechanical relations, retaining the same core representation. Deliverables include annotated test cases, specifications for principle and verification records, reproducible implementations, and research publications. Expansion to additional domains or repair-model training will depend on the results of the core studies.

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